

Perfect Competition

- 1) Perfect competition is an industry with
- A) a few firms producing identical goods.
 - B) many firms producing goods that differ somewhat.
 - C) a few firms producing goods that differ somewhat in quality.
 - D) many firms producing identical goods.
- 2) In a perfectly competitive industry, there are
- A) many buyers and many sellers.
 - B) many sellers, but there might be only one or two buyers.
 - C) many buyers, but there might be only one or two sellers.
 - D) one firm that sets the price for the others to follow.
- 3) In perfect competition, restrictions on entry into an industry
- A) do not exist.
 - B) apply to labor but not to capital.
 - C) apply to both capital and labor.
 - D) apply to capital but not to labor.
- 4) In perfect competition,
- A) there are significant restrictions on entry.
 - B) each firm can influence the price of the good.
 - C) there are few buyers.
 - D) all firms in the market sell their product at the same price.
- 5) Economists assume that a perfectly competitive firm's objective is to maximize its
- A) revenue.
 - B) economic profit.
 - C) output price.
 - D) quantity sold
- 6) Total economic profit is
- A) total revenue minus total cost.
 - B) marginal revenue minus marginal cost.
 - C) total revenue divided by total cost.
 - D) marginal revenue divided by marginal cost.

7) The economic profit of a perfectly competitive firm

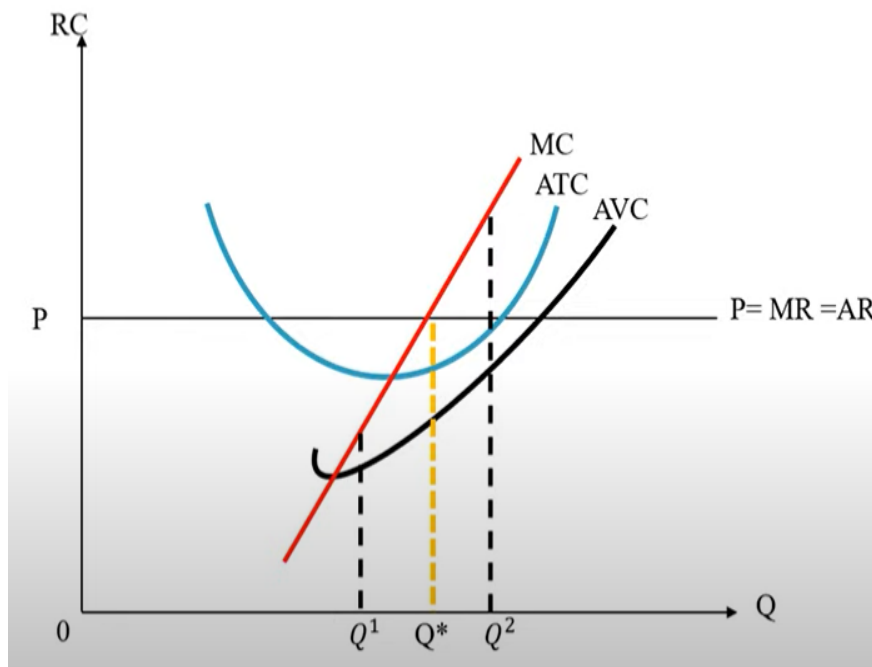
A) is less than its total revenue.

B) is greater than its total revenue.

C) equals its total revenue.

D) is less than its total revenue if its supply curve is inelastic and is greater than its total revenue if its supply curve is elastic.

8)



From the figure above, identify the optimum level of output and explain your choice.

Ans 8. Q^* is the optimum level of output here. At this point, $MR=MC$, which ensures the highest level of profit.

[Q_1 is not the right choice here because at this level of output MR is greater than MC which leaves room for firms for further production as profits will be generated.

Q_2 is not the right choice here because at this level of output MR is less than MC and any further production by the firm will result in losses.]

9. Bob produces DVD movies for sale which requires only a building and a machine that copies the original movies into a DVD. Bob rents a building for \$30,000 per month and rents a machine for \$20,000 a month. These are his fixed costs. His variable costs are given in the table. Current price of a DVD is \$20.

Q	FC	VC	TC= FC+VC	MC= Change in TC/Change in Q	AFC= TFC/Q	AVC= TVC/Q	ATC= TC/Q	TR= PxQ	MR= Change in TR/Change in Q	MR-MC
0	50,000	0	50,000	-	-	-	-	-	-	-
1,000	50,000	5,000	55,000	5	50	5	55	20,000	20	15
2,000	50,000	8,000	58,000	3	25	4	29	40,000	20	17
3,000	50,000	9,000	59,000	1	16.7	3	19.7	60,000	20	19
4,000	50,000	14,000	64,000	5	12.5	3.5	16	80,000	20	15
5,000	50,000	20,000	70,000	6	10	4	14	1,00,000	20	14
6,000	50,000	33,000	83,000	13	8.3	5.5	13.8	1,20,000	20	7
7,000	50,000	49,000	99,000	16	7.1	7	14.1	1,40,000	20	4
8,000	50,000	72,000	1,22,000	23	6.3	9	15.3	1,60,000	20	-3
9,000	50,000	99,000	1,49,000	27	5.6	11	16.6	1,80,000	20	-7
10,000	50,000	1,50,000	2,00,000	51	5	15	20	2,00,000	20	-31

- a. Calculate TC, MC, AFC, AVC, ATC, TR, MR, and [MR-MC] for each level of output.

Ans a. See table

- b. Identify the optimum level of output and explain your answer.

Ans b. 7,000 is the optimum level of output. Here, the difference between MR & MC is the lowest.

- c. What is Bob's TR at the profit maximising level of output?

Ans c. Bob's TR at the profit maximising level of output = $P \times Q = 20 \times 7000 = 1,40,000$

- d. Identify Bob's break-even price and shut down price.

Ans d. Break-even price is where $P = \min ATC$. Hence, the break-even price is \$13.8

Shutdown price is where $P = \min AVC$. Hence, the shutdown price is \$3.

- e. What is Bob's profit at the optimum level of output?

Ans e. Bob's profit at the optimum level of output, $P = TR - TC = 1,40,000 - 99,000 = \$41,000$.

- f. Suppose the price becomes \$15.

I. What will be the new MR?

II. Calculate $[MR - MC]$.

III. Identify the new profit maximizing level of output.

IV. Calculate the new TR.

Ans f (I). New MR = \$15.

Ans f (II).

Q	MC	MR	MR-MC
0	-	-	-
1,000	5	15	10
2,000	3	15	12
3,000	1	15	14
4,000	5	15	10
5,000	6	15	9
6,000	13	15	2
7,000	16	15	-1
8,000	23	15	-8
9,000	27	15	-12
10,000	51	15	-36

Ans f (III). 6,000 is the optimum level of output. Here, the difference between MR & MC is the lowest.

Ans f (IV). New TR = $P \times Q = 15 \times 6000 = \$90,000$

- g. If price continues to stay at \$15, should Bob continue producing DVD movies or leave the business in the short run?

Ans g. A price of \$15 is greater than the min ATC (\$13.8) and the min AVC (\$3). This means both FC and VC will be covered and Bob will make a profit. Hence, Bob should continue producing DVD movies.

Note: All these problems have been taken from external sources.